

Searching for Parallel Trends: Discovering Diff-in-Diff Estimators

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Basic setup

Data is observed for a set of units over three time periods

- In the first two periods, no units are treated yet
- In the third period, one group of units receives treatment, while the rest remain untreated

Parallel trends assumption (PTA): outcome trends would have been the same for both groups in the absence of treatment

Identification: Under PTA, the control group's trend can be used as proxy for treated group counterfactual trend

Example: impacts of a minimum wage increase

Classic reference is Card and Krueger (1994)

- Studies the impact of a minimum wage increase in New Jersey (NJ) on employment and wages
- Pennsylvania (PA) used as a control group: similar related economies and geographic proximity makes PTA more plausible

Relevant literature

Target: conditional average treatment effects on the treated (CATT)

- Subgroup treatment effects defined in terms of time-invariant features of the unit (company, individual, state, etc)

Problem: PTA condition does not inherently survive aggregation and disaggregation (across features)

- PTA may hold marginally but not conditionally and vice versa
- Estimating conditional effects requires scrutinizing the validity of PTA for each candidate subgroup individually
- If the primary goal is to uncover novel subgroups, such a case-by-case approach is infeasible.

This paper

Contribution

We propose an algorithm for estimating CATT which extracts (approximately) PTA-valid subgroups based on an analysis of pre-trend data.

Model and notation

For treatment groups $g \in \{0, 1\}$ and periods $t \in \{-1, 0, 1\}$, outcome Y , time-invariant features X , treatment indicator Z , and unobserved errors U :

$$Y_g = \mu_g(X) - \gamma(X)D_{-1} + \beta_g(X)D_1 + \tau_g(X)Z + \alpha_g(X)ZD_1 + \epsilon_g$$

$$D_{t'} = \mathbf{1}\{t = t'\}$$

$$\epsilon_g = f_g(X, Z, t, U) - \mathbb{E}(f_g(X, Z, t, U) \mid X, Z, t)$$

Target estimand

Let $\mathcal{X}$ denote the support of X , and $\mathbb{E}_{g'}$ denote an expectation taken with respect to the distribution $X \mid g = g'$

Target estimand

$$\begin{aligned} CATT(x) &:= \mathbb{E}_1[f_1(X, Z = 1, t = 1, U) - f_1(X, Z = 0, t = 1, U) \mid X = x] \\ &= \tau_1(x) + \alpha_1(x). \end{aligned}$$

Problem: $\mathbb{E}_1[f_1(X, Z = 0, t = 1, U) \mid X = x]$ is not observed

DiD estimand

$$\begin{aligned}\Delta_1(x) &:= \mathbb{E}_1[f_1(X, Z = 1, t = 1, U) - f_1(X, Z = 0, t = 0, U) \mid X = x] \\ &\quad - \mathbb{E}_0[f_0(X, Z = 0, t = 1, U) - f_0(X, Z = 0, t = 0, U) \mid X = x] \\ &= (\mathbb{E}_1[Y_1 \mid X = x, Z = 1, t = 1] - \mathbb{E}_1[Y_1 \mid X = x, Z = 0, t = 0]) \\ &\quad - (\mathbb{E}_0[Y_0 \mid X = x, Z = 0, t = 1] - \mathbb{E}_0[Y_0 \mid X = x, Z = 0, t = 0]).\end{aligned}$$

Direct substitution shows that:

$$\Delta_1(x) = \underbrace{[\beta_1(x) - \beta_0(x)]}_{\text{Bias}} + \underbrace{\tau_1(x) + \alpha_1(x)}_{\text{CATT}(x)}. \quad (1)$$

Conditional PTA

There exists some $S \subseteq \mathcal{X}$ such that:

$$\mathbb{E}_1[\beta_1(X) \mid X \in S] = \mathbb{E}_0[\beta_0(X) \mid X \in S]. \quad (2)$$

It follows from PTA on S that:

$$\mathbb{E}[\Delta_1(X) \mid X \in S] = \mathbb{E}_1[\tau_1(X) + \alpha_1(X) \mid X \in S] \quad (3)$$

The role of pre-treatment data

Portability of zero-bias regions

For a given $S \in \mathcal{X}$, we say that portability holds if:

$$\begin{aligned} \mathbb{E}_1[\gamma_1(X) \mid X \in S] - \mathbb{E}_0[\gamma_0(X) \mid X \in S] &= 0 \\ \rightarrow \mathbb{E}_1[\beta_1(X) \mid X \in S] - \mathbb{E}_0[\beta_0(X) \mid X \in S] &= 0. \end{aligned} \tag{4}$$

Under this assumption, we can ‘test’ whether PTA holds in a certain region, or gauge the amount of bias in the DiD estimand

CATT at finest PTA region

Let $S(x)$ denote the finest PTA region to which point x belongs. For every $x \in \mathcal{X}$, our target is:

$$CATT(S(x)) := \mathbb{E}_1[\tau_1(X) + \alpha_1(X) \mid X \in S(x)]. \quad (5)$$

Because our target estimand is a very fine one, we develop an algorithm to search for small neighborhoods around a given x where PTA holds

Algorithm

- 1 Construct an initial partition of the dataset
- 2 Define a target point x^*
- 3 Let $S_0(x^*)$ denote the region where x^* falls into in the initial partition. Select a region $S_0^*(x^*)$ among *contiguous* regions of $S_0(x^*)$ in the partition
- 4 If

$$|\mathbb{E}_1[\hat{\gamma}_1(X) \mid X \in S_0(x^*) \cup S_0^*(x^*)] - \mathbb{E}_0[\hat{\gamma}_0(X) \mid X \in S_0(x^*) \cup S_0^*(x^*)]| < |\mathbb{E}_1[\hat{\gamma}_1(X) \mid X \in S_0(x^*)] - \mathbb{E}_0[\hat{\gamma}_0(X) \mid X \in S_0(x^*)]|,$$

update $S_1(x^*) \rightarrow S_0(x^*) \cup S_0^*(x^*)$; otherwise, update $S_1(x^*) \rightarrow S_0(x^*)$

Algorithm illustration

Algorithm

The search proceeds for a pre-specified number of iterations m . Let $\theta_1 = (\beta_1 + \alpha_1 + \tau_1)$:

- If, by the end of the search,

$$|\mathbb{E}_1[\hat{\gamma}_1(X) \mid X \in S_m(x^*)] - \mathbb{E}_0[\hat{\gamma}_0(X) \mid X \in S_m(x^*)]| < \epsilon,$$

for a pre-specified ϵ , set

$$\hat{CATT}(S(x^*)) \rightarrow \mathbb{E}_1[\hat{\theta}_1(X) \mid X \in S_m(x^*)] - \mathbb{E}_0[\hat{\beta}_0(X) \mid X \in S_m(x^*)];$$

- Otherwise, set

$$\hat{CATT}(S(x^*)) \rightarrow k \sim \mathcal{N}(0, \text{sd}(y)).$$

Posterior estimates of $\mathbb{E}[Y \mid X, Z, g, t]$ obtained by BART (Chipman et al., 2010), as implemented in the *stochtree* package¹

- We fit the trees on X with leaf-level regressions of the form

$$\mu_0(1 - g) + \mu_1g - \gamma_0(1 - g)D_{-1} - \gamma_1gD_{-1} + \beta_0(1 - g)D_1 + \theta_1gD_1.$$

- $CATT(S(x))$ posterior draws obtained by applying our search procedure to each posterior draw of $(\hat{\gamma}_0(x), \hat{\gamma}_1(x), \hat{\beta}_0(x), \hat{\theta}_1(x))$ obtained by BART

¹<https://stochtree.ai/>

We consider two additional estimators for comparison

- $CATT(x)$: the pointwise DiD estimator
- $ATT(x)$: assume no heterogeneity and take the CATT for each point as the global ATT

Data-generating processes

We consider a single discrete feature X with support $\mathcal{X} = \{1, \dots, 16\}$, generated for each group from a latent variable U_g . Importantly, we set the CATT as a step function:

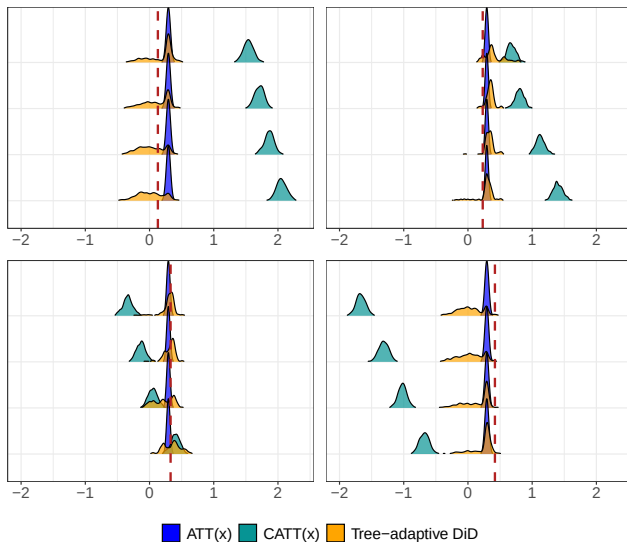
$$\tau(x) = \begin{cases} 0.1 & \text{if } x \in [1, 4] \\ 0.17 & \text{if } x \in [5, 8] \\ 0.25 & \text{if } x \in [9, 12] \\ 0.32 & \text{if } x \in [13, 16] \end{cases} \quad \alpha(x) = \begin{cases} 0.03 & \text{if } x \in [1, 4] \\ 0.06 & \text{if } x \in [5, 8] \\ 0.08 & \text{if } x \in [9, 12] \\ 0.1 & \text{if } x \in [13, 16]. \end{cases} \quad (6)$$

Data-generating processes

We consider three cases:

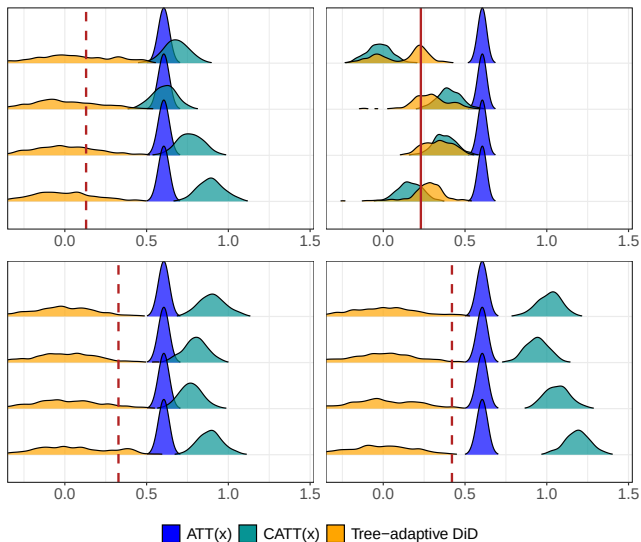
- 1 **PTA holds only aggregated over $\mathcal{X}$ (global PTA):** a scenario where the $ATT(x)$ estimator should perform well since CATT heterogeneity is not very strong
- 2 **PTA holds only aggregated over $x \in \{5, 6, 7, 8\}$ (local PTA):** a scenario where our local estimator should be able to detect the region where PTA holds and where it does not, flagging the latter as unidentified, and producing better CATT estimates for the former
- 3 **PTA holds for all $x \in \mathcal{X}$ (pointwise PTA):** a scenario where $CATT(x)$ should perform well

Results I



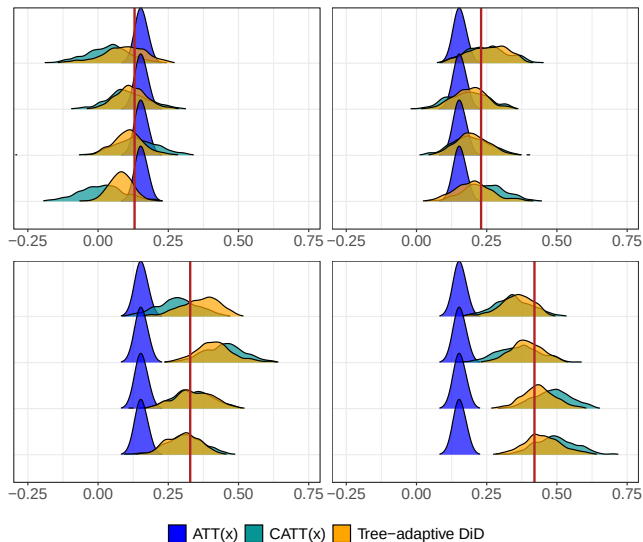
(a) Global PTA

Results II



(b) Local PTA

Results III



(c) Pointwise PTA

Summary of results

- Each of the alternative estimators only performs well in the case where PTA holds at the right level (globally or pointwise)
- $CATT(S(x))$ performs well in all scenarios
- In the local PTA case, $CATT(S(X))$ is able to pick out the right reason, and defaults to the prior almost completely in the non-PTA regions

Contributions

- Formalize issues when learning covariate-based heterogeneity of treatment effects in DiD setup
- Formalize some of the ideas behind pre-trend tests
- Framing pre-trend tests as an exploration of a posterior rather than a pre-estimation procedure allows one to avoid the limitations of typical pre-test procedures

Limitations/room for extensions

- Only time-invariant features
- Single treatment period

- Abadie, A. (2005). Semiparametric difference-in-differences estimators. *The review of economic studies*, 72(1):1–19.
- Ashenfelter, O. and Card, D. (1985). Using the longitudinal structure of earnings to estimate the effect of training programs. *The Review of Economics and Statistics*, pages 648–660.
- Athey, S. and Imbens, G. W. (2022). Design-based analysis in difference-in-differences settings with staggered adoption. *Journal of Econometrics*, 226(1):62–79.
- Bilinski, A. and Hatfield, L. A. (2018). Nothing to see here? non-inferiority approaches to parallel trends and other model assumptions. *arXiv preprint arXiv:1805.03273*.
- Callaway, B. and Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of econometrics*, 225(2):200–230.

Bibliography II

- Card, D. and Krueger, A. (1994). Minimum wages and employment: A case study of the fast-food Industry in New Jersey and Pennsylvania. *The American Economic Review*, 84(4):772–793.
- Chipman, H. A., George, E. I., and McCulloch, R. E. (2010). BART: Bayesian additive regression trees.
- De Chaisemartin, C. and d'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American economic review*, 110(9):2964–2996.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of econometrics*, 225(2):254–277.
- Heckman, J. J., Ichimura, H., and Todd, P. (1998). Matching as an econometric evaluation estimator. *The review of economic studies*, 65(2):261–294.

Bibliography III

- Kahn-Lang, A. and Lang, K. (2020). The promise and pitfalls of differences-in-differences: Reflections on 16 and pregnant and other applications. *Journal of Business & Economic Statistics*, 38(3):613–620.
- Nie, X., Lu, C., and Wager, S. (2021). Nonparametric heterogeneous treatment effect estimation in repeated cross sectional designs. In *Handbook of Statistical Methods for Precision Medicine*, pages 181–206. Chapman and Hall/CRC.
- Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3):305–322.
- Sant’Anna, P. H. and Zhao, J. (2020). Doubly robust difference-in-differences estimators. *Journal of econometrics*, 219(1):101–122.
- Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of econometrics*, 225(2):175–199.

Seminal: introduced by Ashenfelter and Card (1985), formalized by Heckman et al. (1998)

Conditional PTA: Abadie (2005) and Sant'Anna and Zhao (2020) target the ATT, Nie et al. (2021) target the CATT

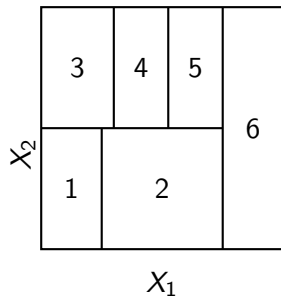
- PTA assumed to hold pointwise

Problems with pre-trend tests: Roth (2022); Bilinski and Hatfield (2018); Kahn-Lang and Lang (2020)

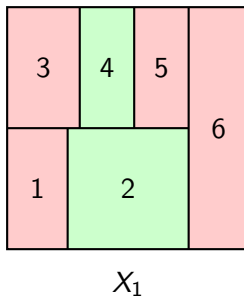
- Low power
- Conditioning on zero pre-trend can induce bias in ATT

Illustration of valid and invalid clusters

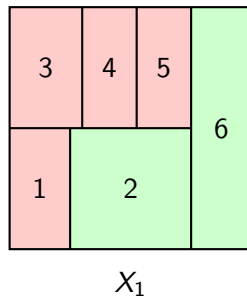
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(a) Initial partition



(b) Invalid partition



(c) Valid partition